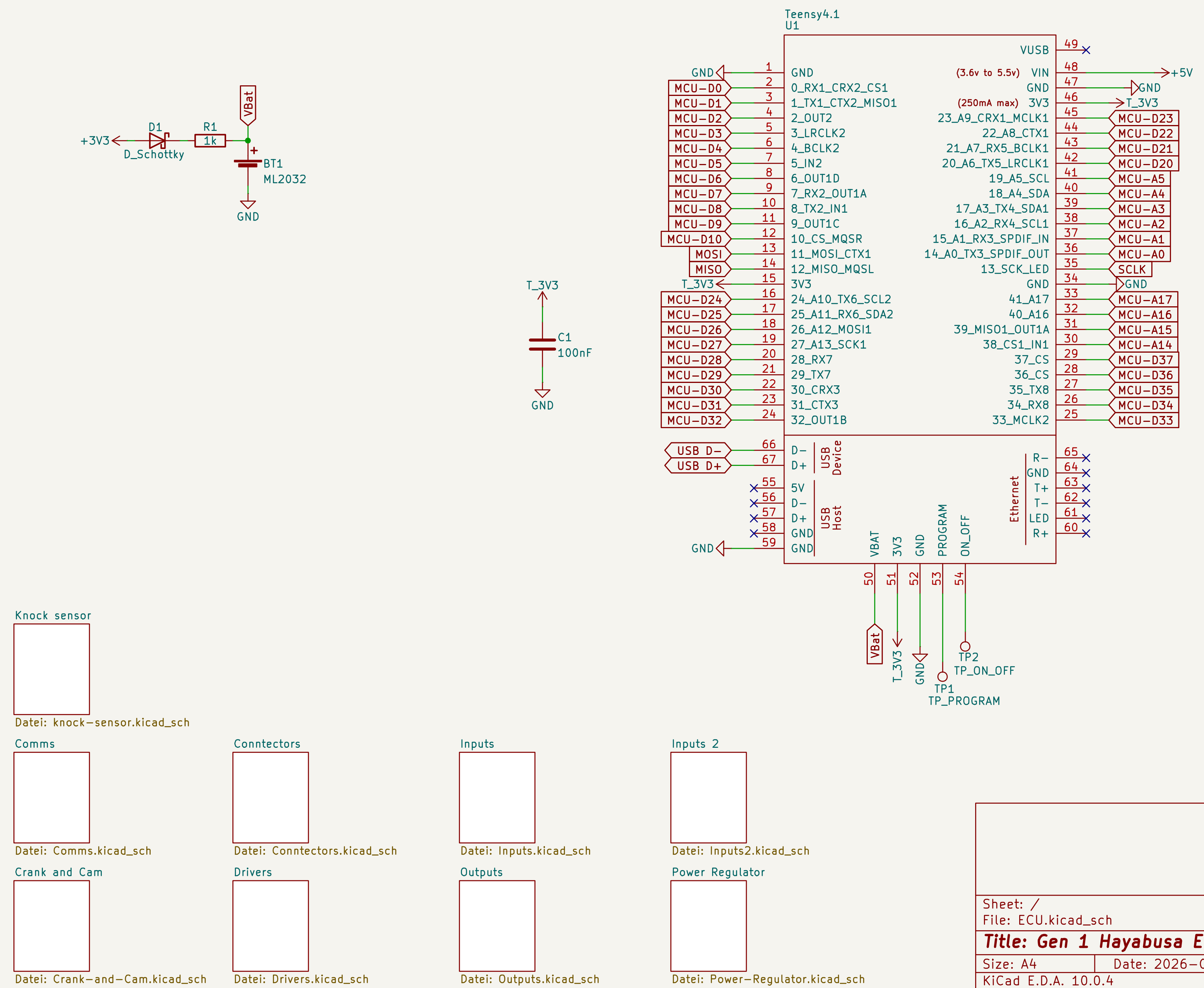

Sheet: /
File: ECU.kicad_sch

Title: Gen 1 Hayabusa ECU

Size: A4 | Date: 2026-08-25 | **Rev: 3.0-draft**

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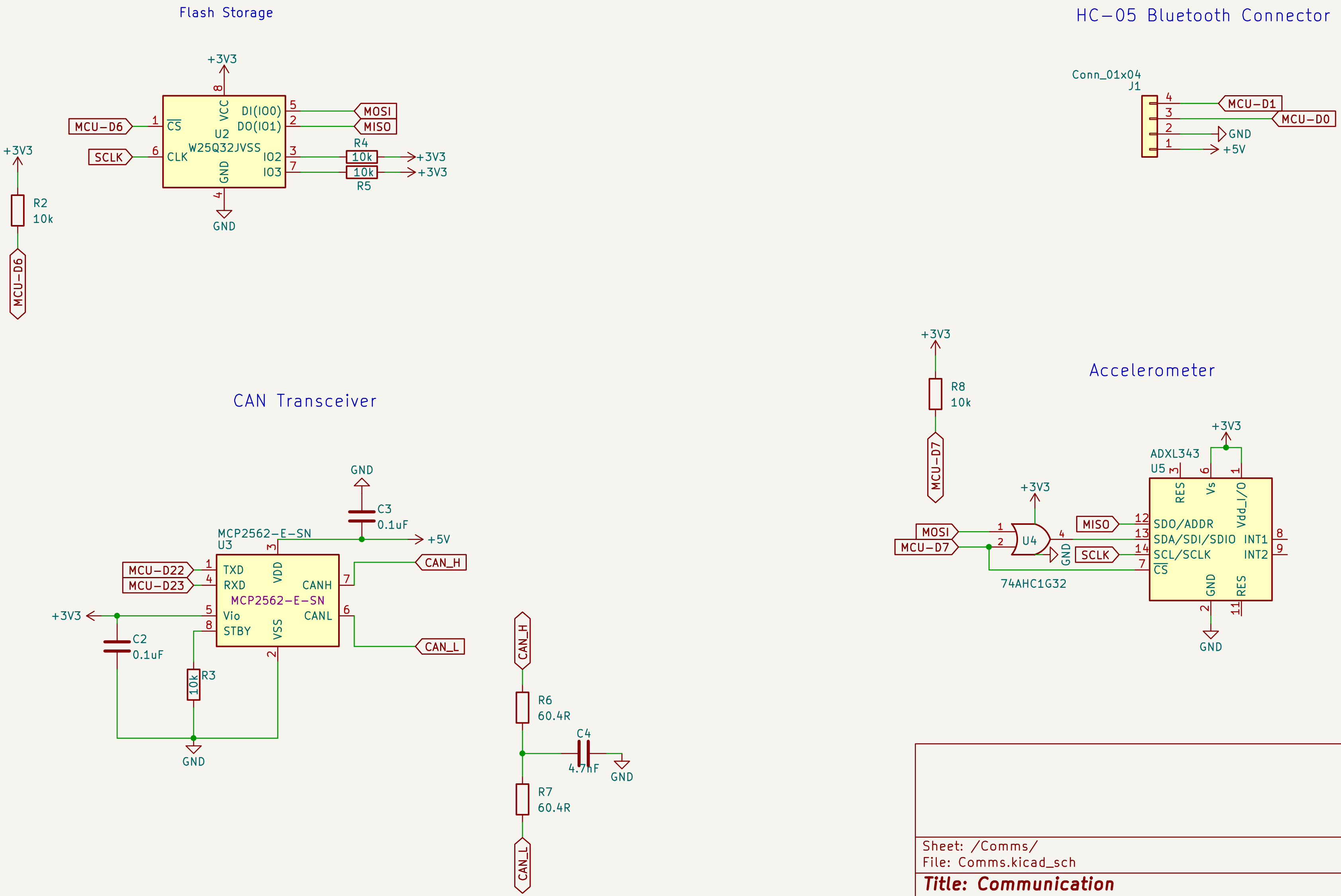

Sheet: /Comms/
File: Comms.kicad_sch

Title: Communication

Size: A4	Date: 2026-08-25	**Rev: 3.0-draft**
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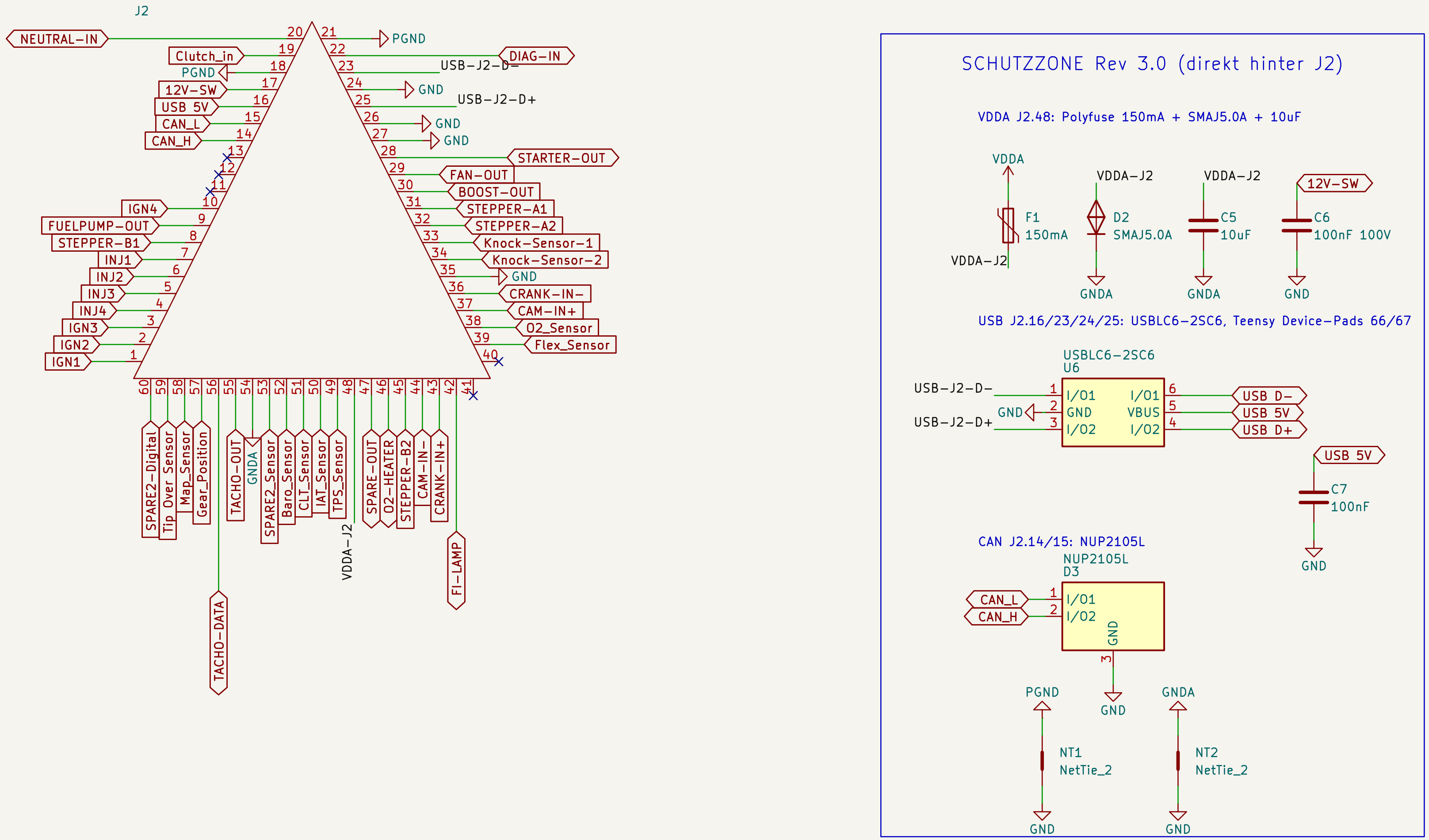

Sheet: /Conntectors/
File: Conntectors.kicad_sch

Title: Connectors

Size: A4 | Date: 2026-08-25 | **Rev: 3.0-draft**

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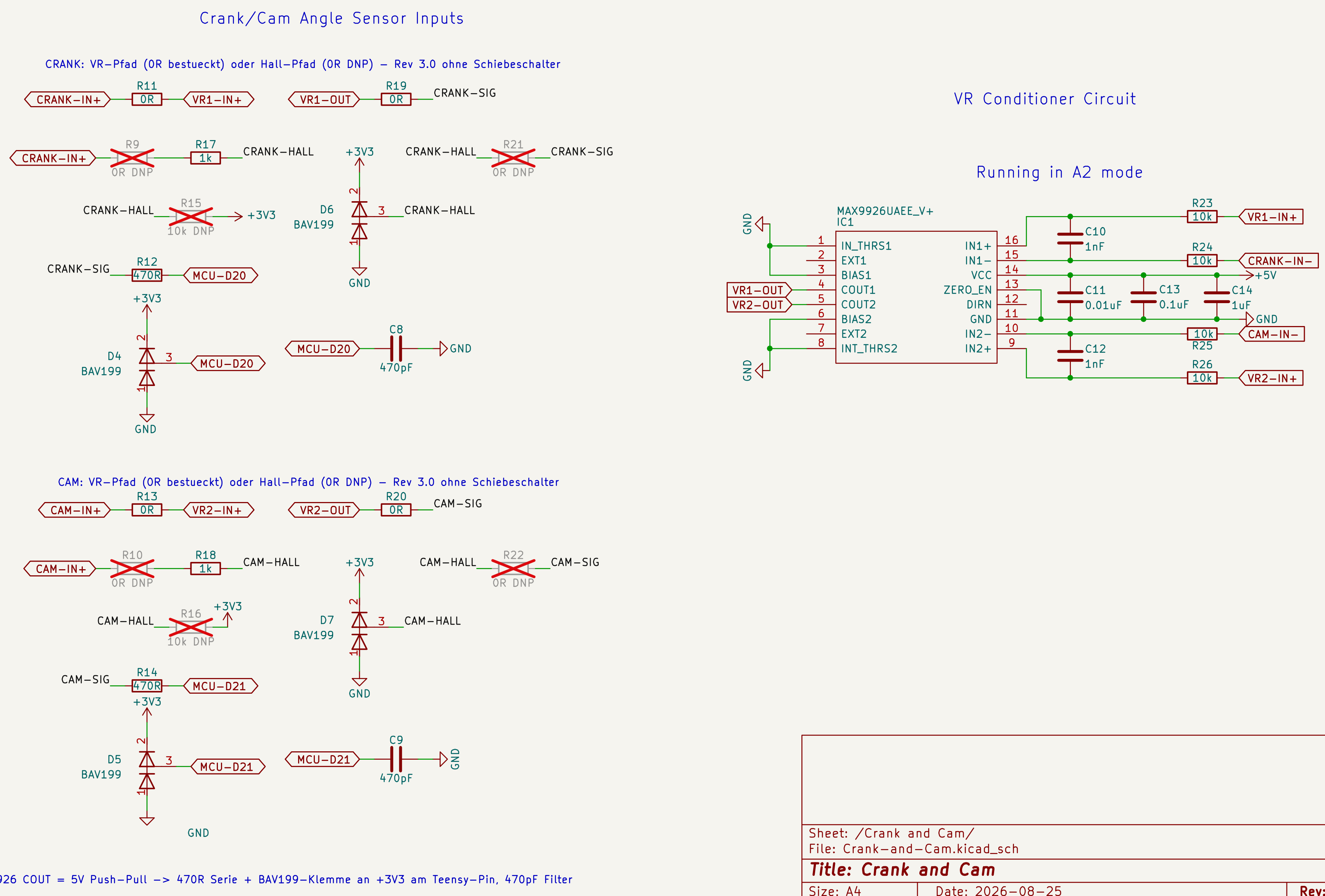
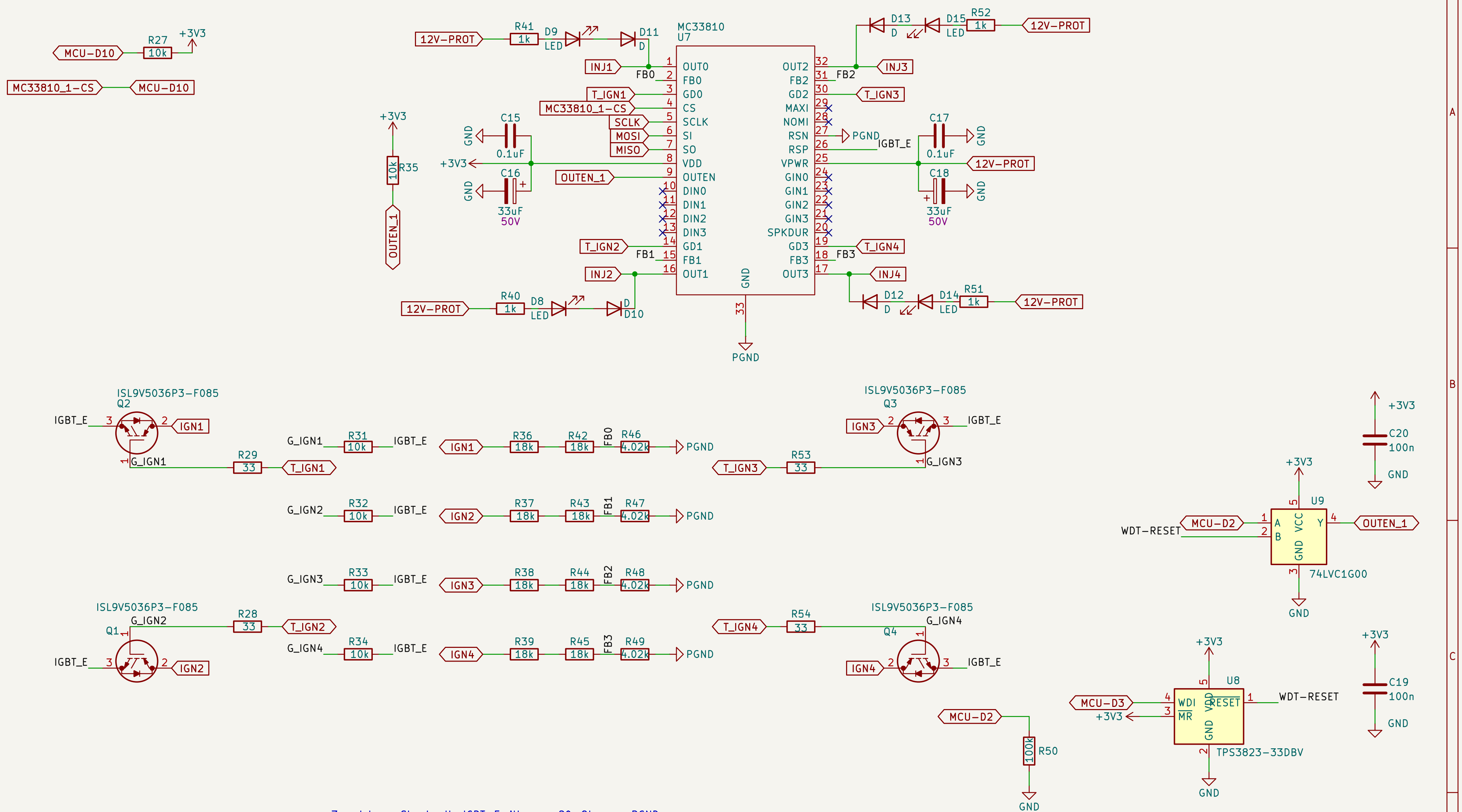


Sheet: /Crank and Cam/
File: Crank-and-Cam.kicad_sch

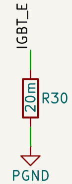
Title: Crank and Cam

Size: A4 | Date: 2026-08-25 | **Rev: 3.0-draft**

KiCad E.D.A. 10.0.4 | Id: 4/10



Zuendstrom−Shunt: alle IGBT−Emitter −> 20mOhm −> PGND
RSP=IGBT_E, RSN=PGND (Kelvin−Layout!), Schwellen per SPI−DAC, Gain=2



*Note: Max current per injector channel is 3.5A

Rev 3.0: MC33810 GD0−3 (5−9V Ladungspumpe) −> 33R −> ISL9V5036 direkt.
TC4424/SW3/F1/IGN_V+ entfallen. VPWR/Freilauf an 12V−PROT. U7−GND = PGND (Injektor−Rueckstrom).

OUTEN−Safety: OUTEN aktiv LOW, R16 Pull−Up +3V3 (Boot=AUS).
Freigabe nur bei MCU−D2=HIGH UND Watchdog−RESET=HIGH (WDI<1.6s von MCU−D3).
Brownout/MCU−Absturz −> RESET low −> NAND=HIGH −> alle 8 Endstufen aus.

Sheet: /Drivers/
File: Drivers.kicad_sch

Title: Ignition and Injector Driver

Size: A4 Date: 2026−08−25 **Rev: 3.0−draft**
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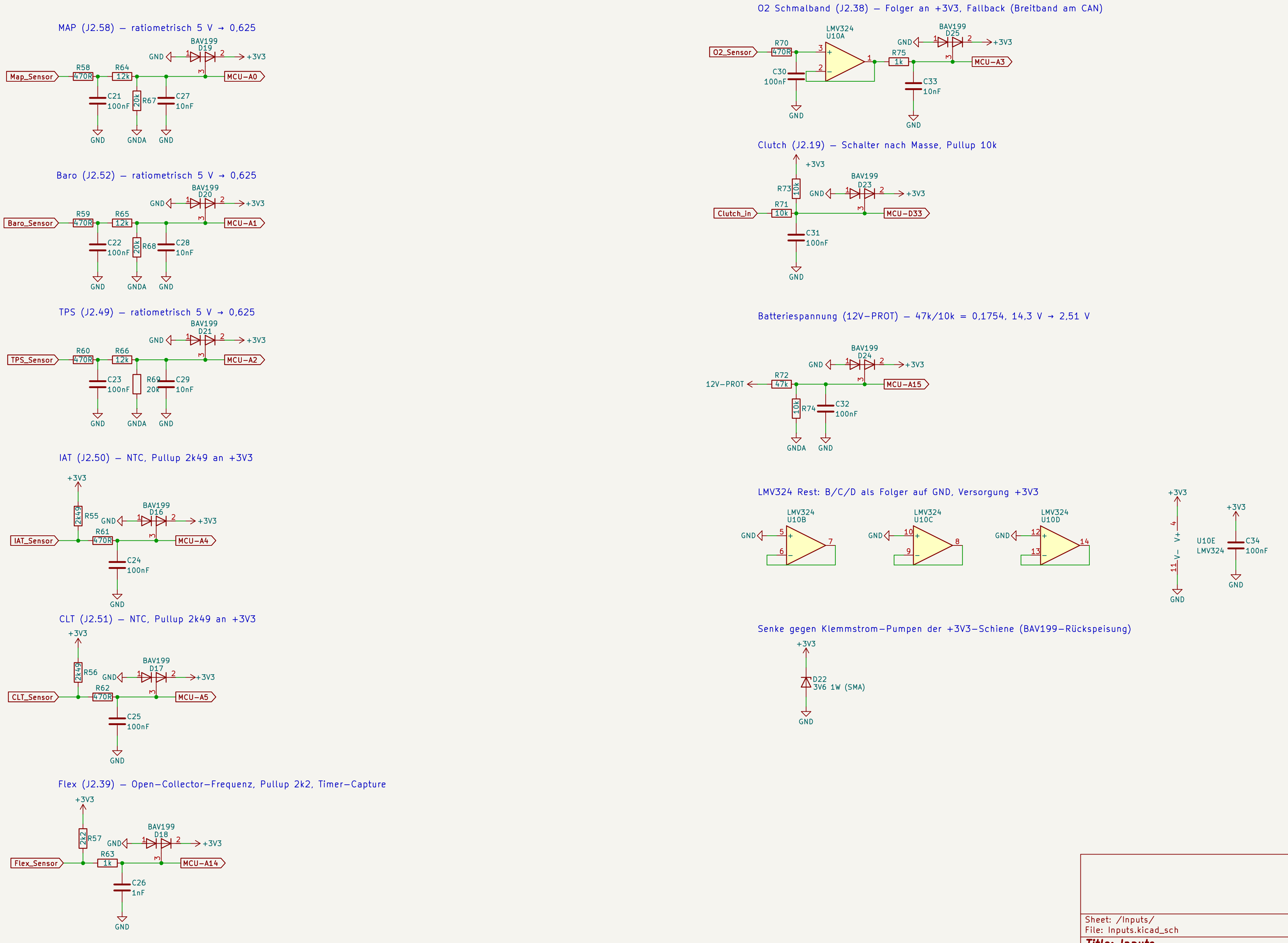

Sheet: /Inputs/
File: Inputs.kicad_sch

Title: Inputs

Size: A3 | Date: 2026-08-25 | Rev: 3.0-draft

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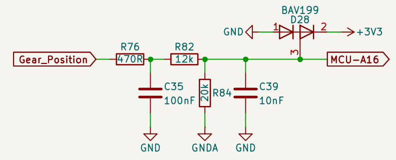
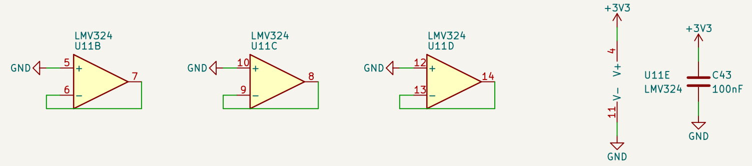
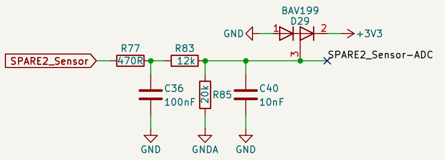
Gangsensor (J2.57) – ratiometrisch 5 V → 0,625



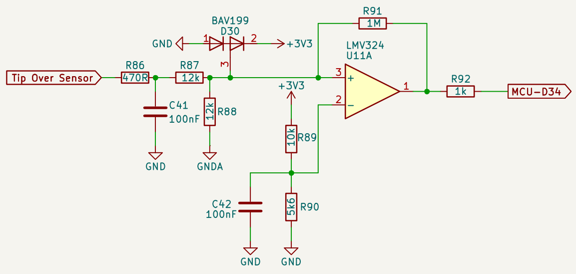
LMV324 Rest: B/C/D als Folger auf GND, Versorgung +3V3



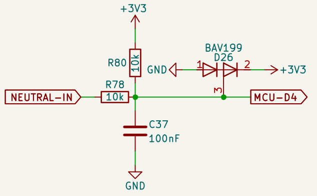
Spare-Analog (J2.53) – Frontend vorbereitet, kein ADC-Pin frei – Option Rev 3.1



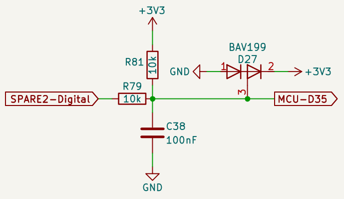
Tip-Over (J2.59) – 0,4-1,4 V normal / 3,7-4,4 V gekippt → Teiler 0,5 → Komparator 1,2 V, Hysterese ~20 mV



Neutral (J2.20) – Schalter nach Masse, Pullup 10k



Spare-Digital (J2.60) – universell, Pullup 10k (für aktiv treibende 5-V-Sensoren: Serien-R + Klemme)



Sheet: /Inputs 2/
File: Inputs2.kicad_sch

Title: Inputs 2

Size: A3 | Date: 2026-08-25 | Rev: 3.0-draft

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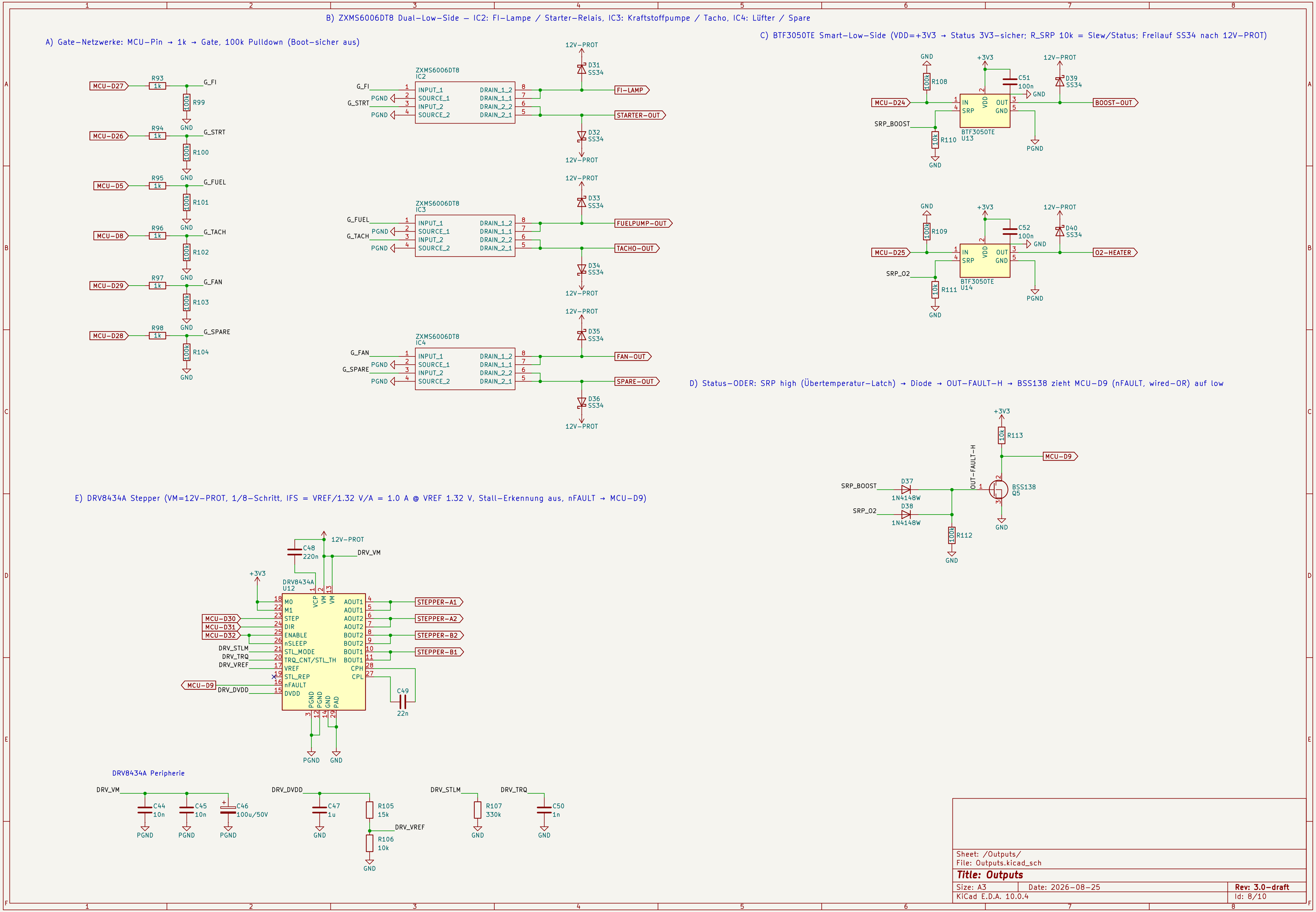
B) ZXMS6006DT8 Dual-Low-Side – IC2: Fl-Lampe / Starter-Relais, IC3: Kraftstoffpumpe / Tacho, IC4: Lüfter / Spare
A) Gate-Netzwerke: MCU-Pin → 1k → Gate, 100k Pulldown (Boot-sicher aus)
C) BTF3050TE Smart-Low-Side (VDD=+3V3 → Status 3V3-sicher; R_SRP 10k = Slew/Status; Freilauf SS34 nach 12V-PROT)
MCU-D27
R93
1k
G_FI
R99
100k
GND
MCU-D26
R94
G_STRT
R100
MCU-D5
R95
G_FUEL
R101
MCU-D8
R96
G_TACH
R102
MCU-D29
R97
G_FAN
R103
MCU-D28
R98
G_SPARE
R104
12V-PROT
D31
SS34
ZXMS6006DT8
IC2
INPUT_1
SOURCE_1
INPUT_2
SOURCE_2
DRAIN_1_2
DRAIN_1_1
DRAIN_2_2
DRAIN_2_1
PGND
FI-LAMP
STARTER-OUT
D32
IC3
D33
D34
FUELPUMP-OUT
TACHO-OUT
IC4
D35
D36
FAN-OUT
SPARE-OUT
GND
+3V3
12V-PROT
R108
C51
100n
D39
MCU-D24
IN
SRP
VDD
OUT
BOOST-OUT
SRP_BOOST
R110
10k
BTF3050TE
U13
PGND
R109
C52
D40
MCU-D25
O2-HEATER
SRP_O2
R111
U14
D) Status-ODER: SRP high (Übertemperatur-Latch) → Diode → OUT-FAULT-H → BSS138 zieht MCU-D9 (nFAULT, wired-OR) auf low
R113
MCU-D9
OUT-FAULT-H
D37
1N4148W
D38
BSS138
Q5
R112
E) DRV8434A Stepper (VM=12V-PROT, 1/8-Schritt, IFS = VREF/1.32 V/A = 1.0 A @ VREF 1.32 V, Stall-Erkennung aus, nFAULT → MCU-D9)
C48
220n
DRV_VM
DRV8434A
U12
M0
M1
STEP
DIR
ENABLE
nSLEEP
STL_MODE
TRQ_CNT/STL_TH
VREF
STL_REP
nFAULT
DVDD
VCP
VM
AOUT1
AOUT2
BOUT2
BOUT1
CPH
CPL
PGND
PAD
MCU-D30
MCU-D31
MCU-D32
DRV_STLM
DRV_TRQ
DRV_VREF
DRV_DVDD
STEPPER-A1
STEPPER-A2
STEPPER-B2
STEPPER-B1
C49
22n
DRV8434A Peripherie
C44
10n
C45
10n
C46
100u/50V
C47
1u
R105
15k
R106
10k
R107
330k
C50
1n
Sheet: /Outputs/
File: Outputs.kicad_sch
Title: Outputs
Size: A3
Date: 2026-08-25
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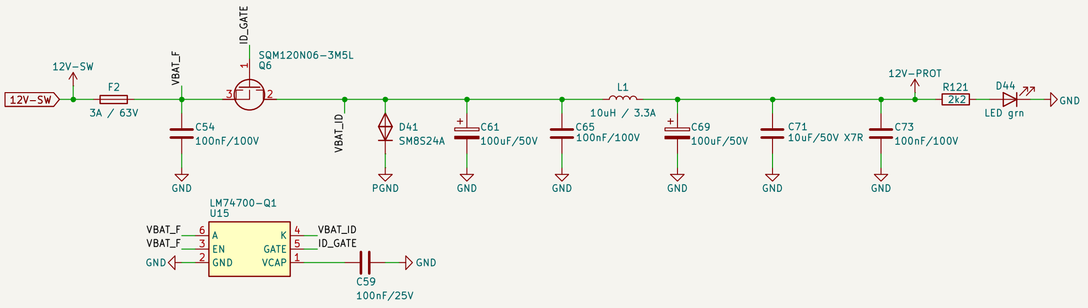
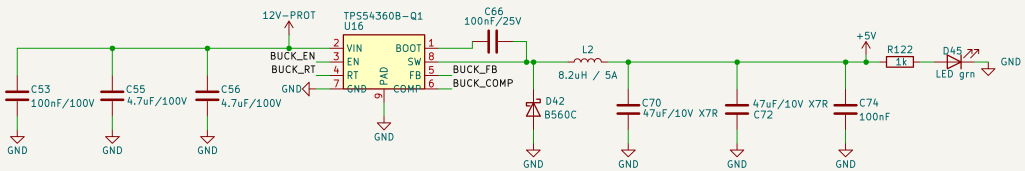
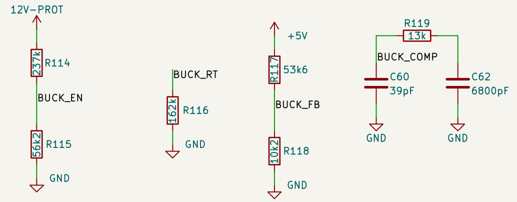
1) EINGANGSSCHUTZ: 12V-SW -> Sicherung -> Ideal-Diode LM74700-Q1 (Verpolschutz, <20 mV) -> SM8S24A (Load Dump ISO 7637-2) -> Pi-Filter -> 12V-PROT



EN = ANODE (immer ein). VCAP 100 nF lt. Datenblatt.

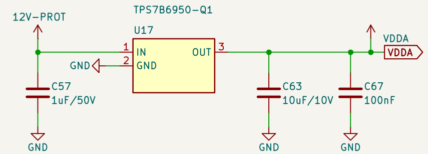
2) +5V HAUPTRAIL: TPS54360B-Q1 Buck, 600 kHz, UVLO 6,0 V ein / 5,2 V aus (Kurbel-Brownout-fest), Werte nach TI-Designbeispiel 5 V (SLVSDV1 Kap. 8.2)



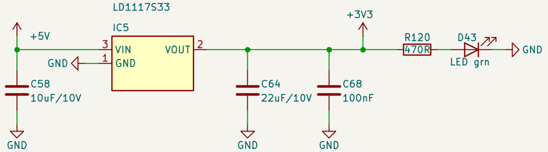


Kompensation/RT/Boot/L/Cout = TI-Designbeispiel 5 V, 600 kHz (R4 13k, C5 6800p, C8 39p, L 8.2uH, Cout 2x47uF, B560C).

3) VDDA SENSORVERSORGUNG: TPS7B6950-Q1 (5 V / 150 mA, kurzschlussfest, 40 V am Eingang). Speist J2.48 + NTC-Pullups; Polyfuse/TVS/10uF am Stecker (Conntectors).



4) +3V3 PERIPHERIE: LD1117S33 aus +5V, alleinige Quelle fuer MC33810-VDD, Flash, ADXL, MAX9926-Logik, LMV324, Watchdog, 74LVC1G125, BTF3050TE-VDD.



Sheet: /Power Regulator/
File: Power-Regulator.kicad_sch

Title: Power Regulator

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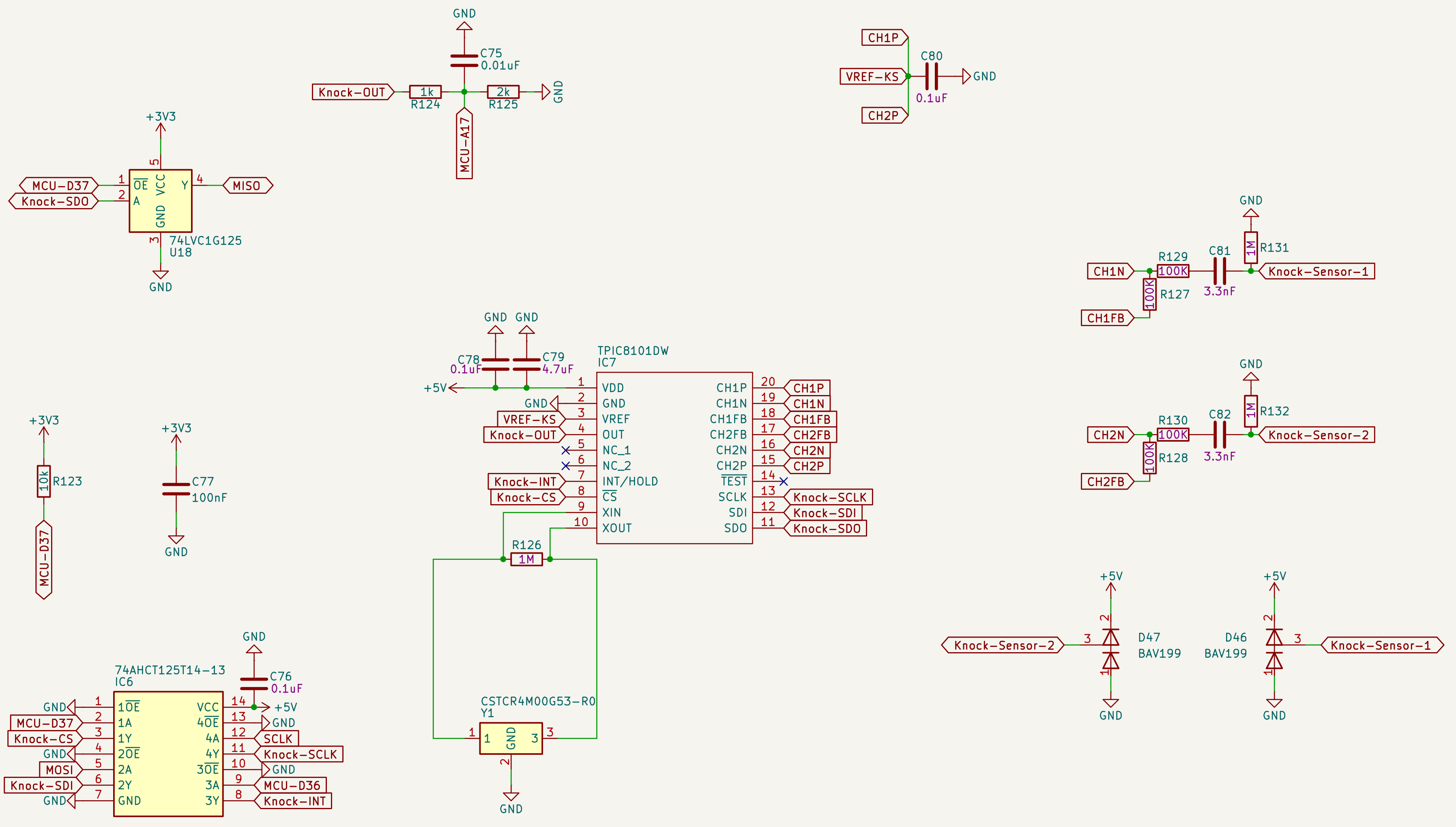

Sheet: /Knock sensor/
File: knock-sensor.kicad_sch

Title: Knock Sensor

Size: A4 | Date: 2026-08-25 | **Rev: 3.0-draft**

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